

Kalen's work when solving the equation $y = 7(x - 5)^2 + 9$ for x is shown. Use Kalen's work to answer questions 1 and 2.

$$y = 7(x - 5)^2 + 9$$

$$y - 9 = 7(x - 5)^2 + 9 - 9$$

$$(y - 9) \div 7 = 7(x - 5)^2 \div 7$$

$$\sqrt{7(y - 9)} = \sqrt{(x - 5)^2}$$

$$\sqrt{7(y - 9)} = x - 5$$

$$\sqrt{7(y - 9)} + 5 = x$$

$$\sqrt{\frac{y - 9}{7}} = \sqrt{(x - 5)^2}$$

$$\sqrt{\frac{y - 9}{7}} = x - 5$$

$$5 + \sqrt{\frac{y - 9}{7}} = x$$

1. Circle the step where Kalen made an error.
2. Correct Kalen's error and the steps that follow to show how to correctly isolate x in the equation. Use the space provided above to show the correct steps.

Solve each equation for the designated variable. Show your work.

3. Solve $y = 4x^2 + px + 40$ for p .

$$y - 40 = 4x^2 + px + 40 - 40$$

$$y - 40 - 4x^2 = 4x^2 + px - 4x^2$$

$$\frac{y - 4x^2 - 40}{x} = \frac{px}{x}$$

$$p = \frac{y - 4x^2 - 40}{x}$$

4. Solve $y = \frac{1}{2}(x - 4)^2 + r$ for r .

$$y - \frac{1}{2}(x - 4)^2 = \frac{1}{2}(x - 4)^2 + r - \frac{1}{2}(x - 4)^2$$

$$y - \frac{1}{2}(x - 4)^2 = r$$

5. Solve $4(x - 2)(x + w) = y$ for w .

$$\frac{4(x - 2)(x + w)}{4(x - 2)} = \frac{y}{4(x - 2)}$$
$$\frac{y}{4(x - 2)} - x = x + w - x$$
$$\frac{y}{4(x - 2)} - x = w$$

6. Solve $y = x^2 - 12x + 3m$ for m .

$$y - (x^2 - 12x) = x^2 - 12x + 3m - (x^2 - 12x)$$
$$\frac{y - x^2 + 12x}{3} = \frac{3m}{3}$$
$$\frac{y - x^2 + 12x}{3} = m$$

7. Solve $y = q(x + 2)^2 - 1$ for q .

$$y + 1 = q(x + 2)^2 - 1 + 1$$
$$\frac{y + 1}{(x + 2)^2} = \frac{q(x + 2)^2}{(x + 2)^2}$$
$$\frac{y + 1}{(x + 2)^2} = q$$

8. Solve $y = a(x - 2)(3x - 4)$ for a .

$$\frac{y}{(x - 2)(3x - 4)} = \frac{a(x - 2)(3x - 4)}{(x - 2)(3x - 4)}$$
$$\frac{y}{(x - 2)(3x - 4)} = a$$

9. Solve $y = -3(x - 2)^2 + 5$ for x .

$$y - 5 = -3(x - 2)^2 + 5 - 5$$
$$\frac{y - 5}{-3} = \frac{-3(x - 2)^2}{-3}$$
$$\sqrt{\frac{y - 5}{-3}} = \sqrt{(x - 2)^2}$$
$$\sqrt{\frac{y - 5}{-3}} = x - 2 \quad \rightarrow \quad 2 + \sqrt{\frac{y - 5}{-3}} = x$$

10. Solve $y = ax^2 + 2x - 9$ for a .

$$y - (2x - 9) = ax^2 + 2x - 9 - (2x - 9)$$
$$\frac{y - 2x + 9}{x^2} = \frac{ax^2}{x^2}$$
$$\frac{y - 2x + 9}{x^2} = a$$